



Girraween High School

2023

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 2

Total Marks: 100

Section 1 (Pages 2 – 5) **10 Marks**

- Attempt Q1 - Q10
- Allow about 15 minutes for this section

General Instructions

- Reading time: 10 minutes
- Working time: 3 Hours
- Write using a black or blue pen
- Board approved calculators may be used
- Laminated reference sheets are provided
- Answer multiple-choice questions by completely colouring in the appropriate circle on your multiple-choice answer sheet.
- Answer questions 11-16 in the appropriate answer booklet and show all relevant mathematical reasoning and/or calculations.

Section 2 (Pages 6-14) **90 marks**

- Attempt Q11 - Q16
- Allow about 2 hours and 45 minutes for this section

Section 1 (10 marks) Multiple choice

Attempt Questions 1-10

Allow about 15 minutes for this section

Question 1

$$i^{2023} =$$

- (A) i (B) -1 (C) $-i$ (D) -1

Question 2

A cubic polynomial $P(x)$ with all real coefficients has roots α, β and γ such that $\alpha^2 + \beta^2 + \gamma^2 < 0$. The equation $P(x) = 0$ has

- (A) No real root. (B) One real root.
(C) Two real roots. (D) Three real roots.

Question 3

One of the roots of the polynomial equation $2z^3 + 7z^2 + 22z - 13 = 0$ is $z = -2 + 3i$. The other two roots are

- (A) $z = -2 - 3i$ and $z = \frac{1}{2}$ (B) $z = -2 - 3i$ and $z = -\frac{1}{2}$
(C) $z = 2 + 3i$ and $z = \frac{1}{2}$ (D) $z = 2 - 3i$ and $z = -\frac{1}{2}$

Question 4

Given that $\underline{p}$ and $\underline{q}$ are non-zero vectors, the contrapositive of:

if $\underline{p} \cdot \underline{q} = 0$ then $\underline{p} \perp \underline{q}$ is

- (A) If $\underline{p} \perp \underline{q}$ then $\underline{p} \cdot \underline{q} = 0$ (B) If $\underline{p}$ is NOT $\perp \underline{q}$ then $\underline{p} \cdot \underline{q} \neq 0$
(C) If $\underline{p} \perp \underline{q}$ then $\underline{p} \cdot \underline{q} \neq 0$ (D) If $\underline{p}$ is NOT $\perp \underline{q}$ then $\underline{p} \cdot \underline{q} = 0$

Multiple choice continues on the following page

Multiple choice (continued)

Question 5

Given that $\vec{p}$ and $\vec{q}$ are non-zero vectors, the converse of:

if $\vec{p} \cdot \vec{q} = 0$ then $\vec{p} \perp \vec{q}$ is

(A) If $\vec{p} \perp \vec{q}$ then $\vec{p} \cdot \vec{q} = 0$

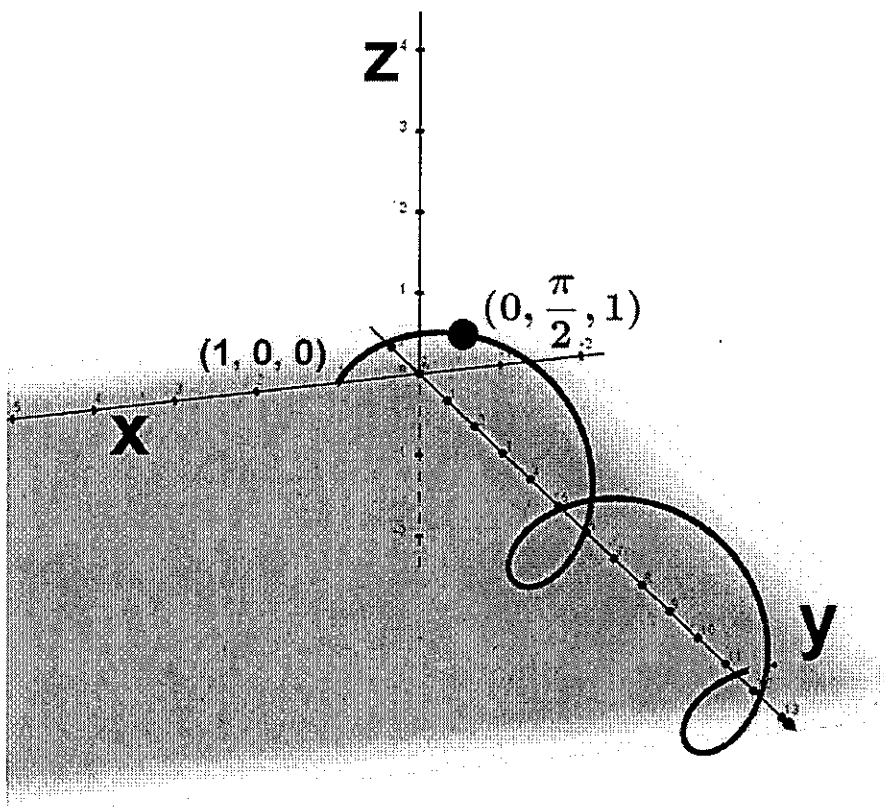
(B) If $\vec{p}$ is NOT $\perp \vec{q}$ then $\vec{p} \cdot \vec{q} \neq 0$

(C) If $\vec{p} \perp \vec{q}$ then $\vec{p} \cdot \vec{q} \neq 0$

(D) If $\vec{p}$ is NOT $\perp \vec{q}$ then $\vec{p} \cdot \vec{q} = 0$

Question 6

A curve in three dimensional space is pictured below.



The equation for this curve is

(A) $(\cos t, \sin t, t)$

(B) $(\cos t, t, \sin t)$

(C) $(t, \cos t, \sin t)$

(D) $(\sin t, t, \cos t)$

Multiple choice continues on the following page

Multiple choice (continued)

Question 7

A particle moving with simple harmonic motion (SHM) starts from rest at $x = 1$. It next stops at $x = 9$. It next returns to $x = 1$ π seconds later. A possible equation for the displacement of this particle at time t is

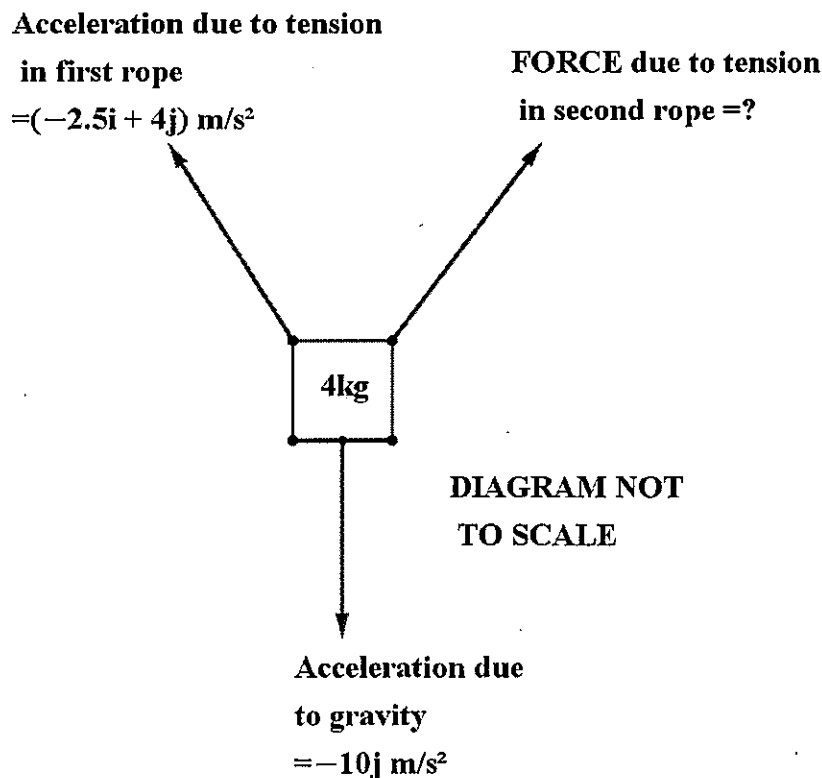
- (A) $x = 4 \sin 2t + 5$ (B) $x = 4 \cos 2t + 5$
(C) $x = -4 \sin 2t + 5$ (D) $x = -4 \cos 2t + 5$

Question 8

A particle with mass 4 kg is hanging stationary from two taut ropes. The *acceleration* due to tension in the first rope (with direction) is $(-2.5\hat{i} + 4\hat{j}) \text{ m/s}^2$. If the acceleration due to gravity is $(-10\hat{j}) \text{ m/s}^2$ (see diagram), the total FORCE in the second rope is

- (A) 10 Newtons (B) 12 Newtons
(C) 13 Newtons (D) 26 Newtons

Diagram for Q8:



Multiple choice continues on the following page

Multiple choice (continued)

Question 9

$$\int_0^a f(2a - x) \cdot dx =$$

(A) $\int_a^{2a} f(x) \cdot dx$

(B) $\int_0^a f(x) \cdot dx$

(C) $\int_{2a}^a f(x) \cdot dx$

(D) $\int_a^0 f(x) \cdot dx$

Question 10

$$\int x^n \ln(x) \cdot dx =$$

(A) $\frac{x^{n+1}}{n+1} - \int x^n \ln x \cdot dx$

(B) $x^{n-1} - \int \frac{x^{n+1}}{n+1} \ln x \cdot dx$

(C) $x^n \ln x - \int \frac{x^n}{n+1} \cdot dx$

(D) $\frac{x^{n+1}}{n+1} \ln x - \int \frac{x^n}{n+1} \cdot dx$

Examination continues on the following page

Section II (90 marks)

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Start the answers to each question on a separate page in your answer booklet.

In Questions 11-16 your responses should include all relevant mathematical reasoning and/ or calculations. (*Note: for complex number questions, the notation cis is acceptable*).

Question 11 (15 marks)

Marks

- | | | |
|------------|--|----------|
| (a) | (i) Find $\frac{1+i\sqrt{3}}{1+i}$. Leave your answer in Cartesian form. | 2 |
| | (ii) Find $\frac{1+i\sqrt{3}}{1+i}$ in modulus/argument form. | 2 |
| | (iii) Hence find the exact value of $\cos \frac{\pi}{12}$. | 1 |
| (b) | (i) By letting $\sqrt{7+24i} = x + iy$, x, y real, find $\sqrt{7+24i}$ in Cartesian form. | 3 |
| | (ii) Hence solve $z^2 + (4-i)z + (2-8i) = 0$. | 2 |
| (c) | Sketch and shade the region in the complex plane where $ z - 1 - i \geq 2$ and $\frac{\pi}{4} \leq \text{Arg } z \leq \frac{\pi}{2}$ hold simultaneously. | 3 |
| (d) | Use DeMoivre's Theorem to find all of the 7 th roots of $128i$.
Leave your answers in modulus/argument form. | 2 |

Examination continues on the following page

Question 12 (15 marks)**Marks**

- (a) Find $\int \frac{1}{2-\cos x} \cdot dx$ 3
- (b) (i) Express $\frac{-16x-64}{(x+3)^2(x-1)}$ in the form $\frac{A}{(x+3)^2} + \frac{B}{(x+3)} + \frac{C}{(x-1)}$. 3
- (ii) Hence find $\int \frac{-16x-64}{(x+3)^2(x-1)} \cdot dx$ 1
- (c) Find $\int x \cos x \cdot dx$ 2
- (d) A particle moves so that its position at time t is given by
 $x = 5 \cos 2t - 5\sqrt{3} \sin 2t$.
- (i) Prove the particle is moving in Simple Harmonic Motion. 1
- (ii) Express the particle's motion in the form
 $x = A \cos (2t + \alpha)$. 2
- (iii) Find the period and the amplitude of the particle's motion
and its initial position. 3

Examination continues on the following page

Question 13**Marks**

- (a) (i) Show BY FINDING IT (NOT BY SUBSTITUTION) that the 4

point of intersection of $\begin{pmatrix} -9 \\ -14 \\ -3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 23 \\ -4 \\ -27 \end{pmatrix} + \lambda_2 \begin{pmatrix} 10 \\ 1 \\ -16 \end{pmatrix}$
is $\begin{pmatrix} 3 \\ -6 \\ 5 \end{pmatrix}$.

- (ii) Show that $\begin{pmatrix} -9 \\ -14 \\ -3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$ is a tangent to the sphere 2

$(x - 1)^2 + (y + 2)^2 + (z - 4)^2 = 21$ and show that the point
of intersection of this line with the sphere is also $\begin{pmatrix} 3 \\ -6 \\ 5 \end{pmatrix}$.

- (iii) Find the equation of the straight line perpendicular to both 3

$\begin{pmatrix} -9 \\ -14 \\ -3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 23 \\ -4 \\ -27 \end{pmatrix} + \lambda_2 \begin{pmatrix} 10 \\ 1 \\ -16 \end{pmatrix}$ passing through $\begin{pmatrix} 3 \\ -6 \\ 5 \end{pmatrix}$.

Hence or otherwise show that $\begin{pmatrix} 23 \\ -4 \\ -27 \end{pmatrix} + \lambda_2 \begin{pmatrix} 10 \\ 1 \\ -16 \end{pmatrix}$ is also a tangent
to the sphere $(x - 1)^2 + (y + 2)^2 + (z - 4)^2 = 21$.

- (b) Prove by mathematical induction that $7^n > 5^n + 2^n \forall n \in \mathbb{Z}^+ \geq 2$. 4

- (c) Prove by contradiction that $\log_7 19$ is irrational. 2

Examination continues on the following page

Question 14 (15 marks)**Marks**

(a) (i) Prove $a^2 + b^2 \geq 2ab \forall$ real a, b . 1

(ii) Prove $a^2 + b^2 + c^2 \geq ab + ac + bc \forall$ real a, b, c . 1

(iii) If a, b, c are the sides of a triangle, prove $c^2 > (a - b)^2$. 1
(Be careful!)

(iv) Hence or otherwise prove that if a, b, c are the sides of a triangle, 1
 $a^2 + b^2 + c^2 < 2(ab + ac + bc)$.

(b) (i) If $I_n = \int_0^{\frac{\pi}{4}} \sin^n x \cdot dx$, show that $I_n = \frac{-\left(\frac{1}{\sqrt{2}}\right)^n}{n} + \frac{n-1}{n} I_{n-2}$. 2

(ii) Hence find $\int_0^{\frac{\pi}{4}} \sin^5 x \cdot dx$ 2

(c) Find $\int \sqrt{\frac{x}{2-x}} \cdot dx$ 3

(d) (i) Use Euler's Theorem to prove that $2i \sin n\theta = e^{ni\theta} - e^{-ni\theta}$. 2

(ii) Hence show that $\sin^5 \theta = \frac{\sin 5\theta}{16} - \frac{5 \sin 3\theta}{16} + \frac{5 \sin \theta}{8}$. 2

Examination continues on the following page

Question 15 (15 marks)**Marks**

- (a) A piston is moving up and down with Simple Harmonic Motion.

It weighs 1.2 kg.

At the top of its motion, the piston is 0.25m above the engine base.

At the bottom of its motion it is 0.15m above the engine base.

The piston completes 100 complete oscillations per second.

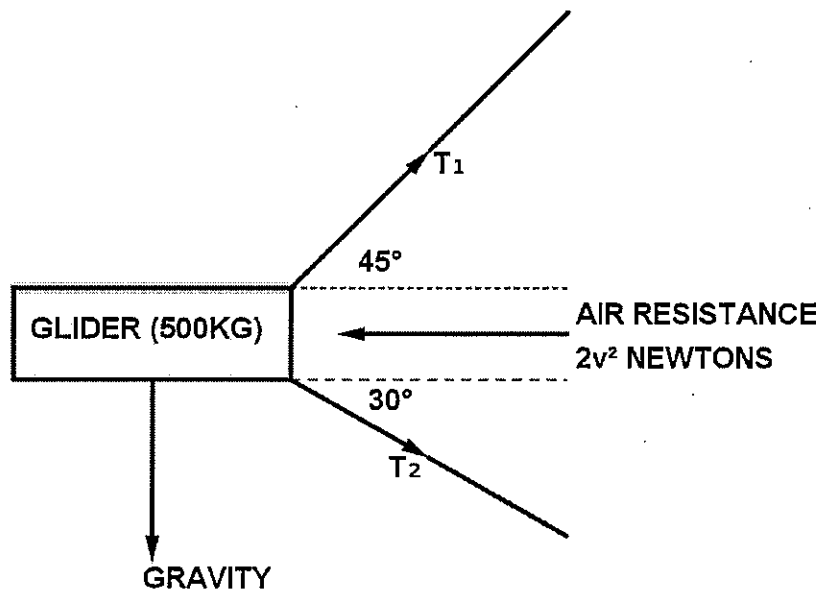
- (i) Use the information above to write the piston's acceleration in the form $\ddot{x} = -n^2(x - C)$. **2**

- (ii) Find the MAXIMUM force the piston exerts in Newtons. **2**

Question 15 continues on the following page

Question 15 (continued)**Marks**

- (b) A 500kg glider is being pulled behind an aeroplane which is flying horizontally. It is attached to the aeroplane by two ropes at angles of 45° above and 30° below the horizontal and the tensions in the two ropes are labelled T_1 and T_2 respectively. It experiences air resistance of $2v^2$ Newtons. The acceleration due to gravity is 10m/s^2 (see diagram below).



- (i) By resolving forces vertically, show that $T_2 = T_1\sqrt{2} - 10\,000$. 2

- (ii) Given $T_1 = 10\,000$ Newtons, by resolving forces horizontally find the total resultant force on the glider. 3

Hence show that $\ddot{x} = 10(\sqrt{2} + \sqrt{6} - \sqrt{3}) - \frac{v^2}{250}$ and find the glider's limiting speed (the speed it approaches but can't reach) in metres per second.

- (iii) Assuming the tensions in the ropes are constant and that the glider's initial speed is 50m/s , find how far the glider travels before it reaches 70m/s . 3

Question 15 continues on the following page

Question 15 (continued)

Marks

- (c) ABC is an equilateral triangle. $\vec{AB} = \vec{p}$ and $\vec{AC} = \vec{q}$.

$\vec{AE} = \vec{FP} = \frac{1}{3} \vec{p}$ and $\vec{AF} = \vec{EP} = \frac{1}{5} \vec{q}$. D is on BC so that

$\vec{BD} = \lambda \vec{BC}$. (So note that D is NOT the midpoint of BC).

(See diagram).

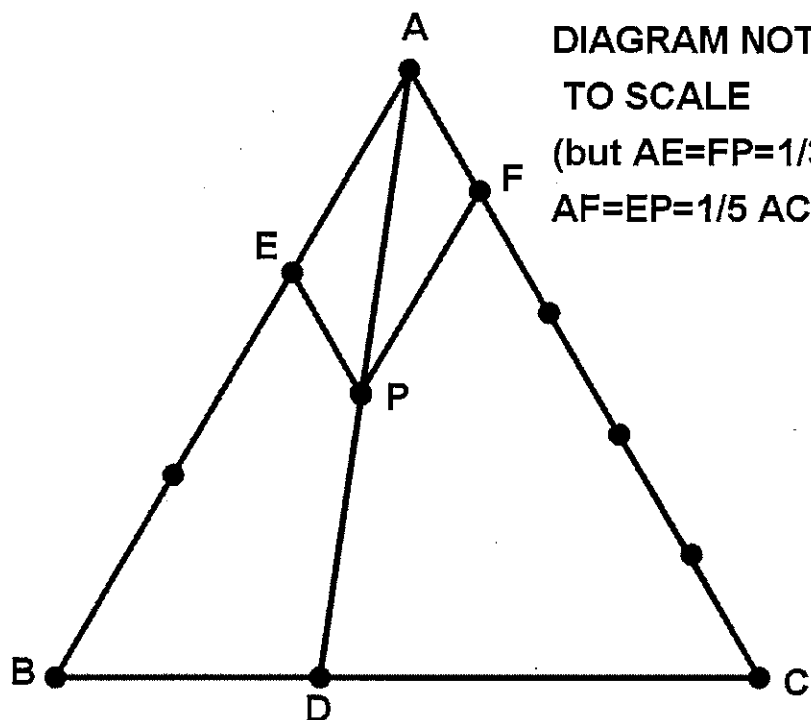


DIAGRAM NOT
TO SCALE

(but $AE=FP=1/3 AB$ and
 $AF=EP=1/5 AC$ as in question)

- (i) Show that $\vec{BD} = \lambda(\vec{q} - \vec{p})$ and $\vec{AD} = \vec{p} + \lambda(\vec{q} - \vec{p})$

1

- (ii) Find the value of λ .

2

Examination continues on the following page

Question 16 (15 marks)**Marks**

- (a) (i) Show using DeMoivre's Theorem that 2
 $\sin 7\theta = 7 \sin \theta - 56 \sin^3 \theta + 112 \sin^5 \theta - 64 \sin^7 \theta.$
- (ii) Hence solve the equation 2
 $64x^7 - 112x^5 + 56x^3 - 7x + 1 = 0.$ Leave your answers
in the form $\sin \theta$, where $0 \leq \theta \leq 2\pi.$
- (iii) Hence or otherwise, show that $\sin \frac{\pi}{14} \sin \frac{5\pi}{14} \sin \frac{17\pi}{14} = -\frac{1}{8}.$ 2
- (b) A projectile with mass M is launched vertically into the air with
initial speed $v_0 m/s$. It experiences air resistance directly against its
motion of $0.2Mv$ Newtons, where v is the current velocity of the
projectile. The ACCELERATION due to gravity is $10m/s^2$. Initially
when the particle is launched the ground is at $x = 0.$
- (i) While the projectile is ascending, show that the time since it 2
was launched is given by $t = -5 \ln \left(\frac{50+v}{50+v_0} \right)$ and find the time taken
to reach the maximum height in terms of $v_0.$
- (ii) Show that $x = -5(50 + v_0)e^{-0.2t} - 50t + 5(50 + v_0)$ and 3
find an expression for the maximum height reached in terms of $v_0.$
- (iii) The particle now starts to fall. It still experiences resistance 2
AGAINST its direction of motion of $0.2Mv$ Newtons and acceleration
due to gravity of $10m/s^2$. Now letting where the projectile starts to
fall be $x = 0$, show that $x = 250e^{-0.2t} + 50t - 250.$
- (iv) If the particle lands back on the ground where it was launched from 2
10 seconds after it was launched, find the initial velocity (v_0) it was
launched at.

END OF EXAMINATION!!!



GIRRAWEE HIGH SCHOOL

MATHEMATICS EXTENSION 2

2023 TRIAL HIGHER SCHOOL CERTIFICATE

*FINAL
Solutions*

Student Number: _____

This Booklet contains the answer sheet for Section 1 and Writing Booklet for Section 2.

Section 1 ANSWER SHEET

Select the alternative A, B, C or D that best answers the question.

1.	A	☐	B	☐	C	☒	D	☐
2.	A	☐	B	☒	C	☐	D	☐
3.	A	☒	B	☐	C	☐	D	☐
4.	A	☐	B	☒	C	☐	D	☐
5.	A	☒	B	☐	C	☐	D	☐
6.	A	☐	B	☒	C	☐	D	☐
7.	A	☐	B	☐	C	☐	D	☒
8.	A	☐	B	☐	C	☐	D	☒
9.	A	☒	B	☐	C	☐	D	☐
10.	A	☐	B	☐	C	☐	D	☒

Instructions

- If you need more paper for Section 2, please ask your supervisor.
- Write your student number on every booklet you use.
- Write on both sides of each sheet of paper.

Total number of booklets used _____.

Solutions - 2023 Girraween HS Ext 2 Trial p.1

Multiple choice:

$$\begin{aligned} (1) & i^{-2023} \\ &= i^{2020} \times i^{-3} \\ &= 1 \times -i \\ &= -i \quad (C) \end{aligned}$$

(2) If $\alpha^2 + \beta^2 + \gamma^2 < 0$
at least 1 of α, β, γ
is not real.

As $P(x)$ has all real
co-efficients, the conjugate
of the non-real root
must also be a root.

$\therefore P(x) = 0$ has 2 non-real roots
& 1 real root. (B)

(3) Like in Q.2, conjugate of
 $-2 + 3i$ [$= -2 - 3i$] is also a root.

By $\alpha + \beta + \gamma = -\frac{b}{a}$

$$\begin{aligned} -2 + 3i + -2 - 3i + \gamma &= -\frac{7}{2} \\ -4 + \gamma &= -\frac{7}{2} \end{aligned}$$

$$\begin{aligned} \gamma &= \frac{1}{2} \\ \text{Roots} &= -2 \pm 3i, \frac{1}{2} \quad (A) \end{aligned}$$

(4) Contrapositive:

If P is NOT $\perp Q$, $P \cdot Q \neq 0$ (B)

(5) Converse: If $P \perp Q$, $P \cdot Q = 0$
(A)

$$(6) t = y = 0 \quad x = \cos 0 = 1 \quad z = \sin 0 = 0$$

$$(1, 0, 0).$$

$$t = y = \frac{\pi}{2} \quad x = \cos \frac{\pi}{2} = 0 \quad y = \sin \frac{\pi}{2} = 1.$$

$$(0, \frac{\pi}{2}, 1).$$

Parametric equation is
 $(\cos t, t, \sin t)$ (B)

(7) Starts at LH end $\rightarrow -\cos$.

Period: $\frac{2\pi}{n} = \pi \Rightarrow n = 2$.

Centre of motion: $x = 5$. Amplitude = 4.

$$x = -4 \cos 2t + 5 \quad (D)$$

(8) Total acceleration of 1st rope
& gravity = $-2.5\hat{i} - 6\hat{j}$

$\therefore$ Total acceleration of 2nd rope
 $= 2.5\hat{i} + 6\hat{j}$.

Magnitude of acceleration
 $= \sqrt{2.5^2 + 6^2}$
 $= 6.5$
By $F = ma$, force = $6.5 \times 4 = 26 \text{ N}$ (D)

(9) Let $u = 2a - x$. $du = -1 \cdot dx$

$$\int_0^a f(2a - x) \cdot dx$$

$$= \int_a^0 f(2a - x) \cdot -1 \cdot dx$$

$$= \int_{2a-a}^{2a-0} f(u) \cdot du$$

$$= \int_a^{2a} f(u) \cdot du = \int_a^{2a} f(x) \cdot dx \quad (A)$$

(10) By $\int uv' \cdot dx$

$$= uv - \int v u' \cdot dx$$

$$\int x^n \ln x \cdot dx = \frac{x^{n+1}}{n+1} \ln x - \int \frac{x^n}{n+1} \cdot dx \quad (D)$$

Solutions: 2023 GMS Trial p.2

$$Q.(11)(a)(i) \frac{1+i\sqrt{3}}{1+i} \times (1-i)$$

$$= \frac{1}{2} [(1+\sqrt{3}) + i(\sqrt{3}-1)]$$

$$(ii) \frac{1+i\sqrt{3}}{1+i}$$

$$= \frac{2 \operatorname{cis} \left[\frac{\pi}{3} \right]}{\sqrt{2} \operatorname{cis} \left[\frac{\pi}{4} \right]}$$

$$= \sqrt{2} \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right)$$

(iii) Hence, equating reals

$$\sqrt{2} \cos \frac{\pi}{12} = \frac{1+\sqrt{3}}{2}$$

$$\cos \frac{\pi}{12} = \frac{1+\sqrt{3}}{2\sqrt{2}}$$

$$\text{or } \cos \frac{\pi}{12} = \frac{\sqrt{2} + \sqrt{6}}{4}$$

$$(b)(i) x + iy = \sqrt{7+24i}, x, y \text{ real.}$$

$$\therefore (x+iy)^2 = 7+24i$$

$$(x^2 - y^2) + 2ixy = 7 + 24i$$

Equating reals, Equating imaginaries

$$x^2 - y^2 = 7 \quad (1) \quad 2xy = 24$$

$$y = \frac{12}{x} \quad (2)$$

Sub. (2) in (1):

$$x^2 - \frac{144}{x^2} = 7$$

$$x^4 - 7x^2 - 144 = 0$$

$$(x^2 - 16)(x^2 + 9) = 0$$

$$x = \pm 4 \quad \uparrow \text{ as } x \text{ real, } x \neq \pm 3i$$

$$y = \frac{12}{x} = \pm 3$$

$$\sqrt{7+24i} = \pm(4+3i)$$

$$(ii) \text{ Solving } z^2 + (4-i)z + (2-8i) = 0$$

$$z = \frac{-(4-i) \pm \sqrt{(4-i)^2 - 4 \times 1 \times (2-8i)}}{2 \times 1}$$

$$z = \frac{-4+i - \sqrt{7+24i}}{2} \quad \text{or} \quad z = \frac{-4+i + \sqrt{7+24i}}{2}$$

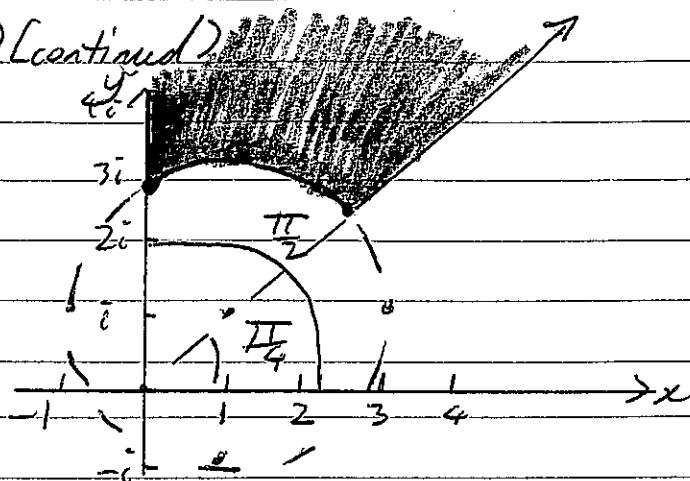
$$z = \frac{-4+i - 4-3i}{2} \quad \text{or} \quad z = \frac{-4+i + 4+3i}{2}$$

$$z = -4-i \quad \text{or} \quad z = 2i$$

Solutions 2023 GHS Trial p. 3

Q.11(i) (continued)

(c)



$$(d) 128i = 128 \operatorname{cis} \frac{\pi}{2}$$

$\therefore \sqrt[7]{128i}$ by De Moivre

$$= 2 \operatorname{cis} \left[\frac{\pi}{14} + \frac{2k\pi}{7}, k=0,1,2,3,4,5,6 \right]$$

$$z = 2 \operatorname{cis} \left[\frac{\pi}{14} \right], 2 \operatorname{cis} \left[\frac{5\pi}{14} \right], 2 \operatorname{cis} \left[\frac{9\pi}{14} \right], 2 \operatorname{cis} \left[\frac{13\pi}{14} \right], \\ 2 \operatorname{cis} \left[\frac{17\pi}{14} \right], 2 \operatorname{cis} \left[\frac{21\pi}{14} \right], 2 \operatorname{cis} \left[\frac{25\pi}{14} \right]$$

or using principal arguments,

$$z = 2 \operatorname{cis} \left[\frac{\pi}{14} \right], 2 \operatorname{cis} \left[\frac{5\pi}{14} \right], 2 \operatorname{cis} \left[\frac{9\pi}{14} \right], 2 \operatorname{cis} \left[\frac{13\pi}{14} \right], \\ 2 \operatorname{cis} \left[-\frac{11\pi}{14} \right], 2 \operatorname{cis} \left[-\frac{\pi}{2} \right], 2 \operatorname{cis} \left[-\frac{3\pi}{14} \right] \\ [= -2i]$$

Solutions: GMSTrial p. 4

$$Q.112)(a) \int \frac{1}{2 - \cos x} dx$$

$$t = \tan\left(\frac{x}{2}\right) \frac{dt}{dx} = \frac{\sec^2(\frac{x}{2})}{2} = \frac{1+t^2}{2}$$

$$dx = \frac{dx}{dt} \cdot dt = \frac{2}{1+t^2} \cdot dt$$

$$= \int \frac{1}{2 - \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} \cdot dt$$

$$= \int \frac{2}{1+3t^2} \cdot dt$$

$$= \frac{2}{3} \int \frac{1}{\frac{1}{3} + t^2} \cdot dt$$

$$= \frac{2\sqrt{3}}{3} \tan^{-1}(t\sqrt{3}) + C$$

$$= \frac{2\sqrt{3}}{3} \tan^{-1}\left[\sqrt{3} \tan\left(\frac{x}{2}\right)\right] + C.$$

$$(b)(i) \frac{-16x - 64}{(x+3)^2(x-1)} = \frac{A}{(x+3)^2} + \frac{B}{(x+3)} + \frac{C}{(x-1)}$$

$$\therefore -16x - 64 = A(x-1) + B(x+3)(x-1) + C(x+3)^2$$

$$\text{Sub. } x=1, -80 = C \times 4^2$$

$$\underline{-5 = C.}$$

$$\text{Sub. } x=-3, -16 = -4A$$

$$\underline{4 = A.}$$

$$\text{Sub. } x=0, A=4, C=-5.$$

$$-64 = -4 - 3B - 45$$

$$\underline{8 = B.}$$

$$\therefore \frac{-16x - 64}{(x+3)^2(x-1)} = \frac{4}{(x+3)^2} + \frac{8}{(x+3)} - \frac{5}{(x-1)}$$

$$(ii) \int \frac{-16x - 64}{(x+3)^2(x-1)} \cdot dx$$

$$= \int \frac{4}{(x+3)^2} + \frac{8}{(x+3)} - \frac{5}{(x-1)} \cdot dx$$

$$= -\frac{4}{(x+3)} + 8 \ln(x+3) - 5 \ln(x-1) + C.$$

Solutions: GHS Trial p.5.

Q.12)(c) $\int x \cos x \cdot dx$ $u = x$ $v = \sin x$
 $u' = 1$ $v' = \cos x$

By $\int u v' \cdot dx = uv - \int v u' \cdot dx$

$\int x \cos x \cdot dx = x \sin x + \int \sin x \cdot dx$

$\int x \cos x \cdot dx = x \sin x + \cos x + C.$

(d)(i) Moving in SHM if $\ddot{x} = -n^2 x$.

$x = 5 \cos 2t - 5\sqrt{3} \sin 2t$

$\dot{x} = -10 \sin 2t - 10\sqrt{3} \cos 2t$

$\ddot{x} = -20 \cos 2t + 20\sqrt{3} \sin 2t$

$= -4(5 \cos 2t - 5\sqrt{3} \sin 2t)$

$\ddot{x} = -4x \Rightarrow$ SHM with $n=2$.

[Could also find $-4x$ separately & do LHS = RHS].

(ii) $5 \cos 2t - 5\sqrt{3} \sin 2t = A \cos 2t \cos \alpha - A \sin 2t \sin \alpha, A > 0$

Equating parts,

$5 \cos 2t = A \cos 2t \cos \alpha$ $-5\sqrt{3} \sin 2t = -A \sin 2t \sin \alpha$

$5 = A \cos \alpha$ (1) $5\sqrt{3} = A \sin \alpha$ (2)

Squaring & adding (1) & (2):

$5^2 + (5\sqrt{3})^2 = A^2(\cos^2 \alpha + \sin^2 \alpha)$

$10 = A.$

Sub. $A=10$ in (1)

$5 = 10 \cos \alpha \Rightarrow \frac{1}{2} = \cos \alpha.$

Sub. $A=10$ in (2)

$5\sqrt{3} = 10 \sin \alpha \Rightarrow \frac{\sqrt{3}}{2} = \sin \alpha.$

$\alpha = \frac{\pi}{3}.$

$x = 10 \cos(2t + \frac{\pi}{3})$

(iii) Period of motion

$= \frac{2\pi}{2}$

$= \pi$ seconds.

Amplitude = 10.

Initial position ($t=0$)

$= 10 \cos \frac{\pi}{3}$

$= x = 5 \text{ m}$ POSITIVE
[right].
of $x=0$.

Solutions: QMS 4U 2023 Trial p. 6

$$(13)(a)(i) \begin{pmatrix} -9 \\ -14 \\ -3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 23 \\ -4 \\ -27 \end{pmatrix} + \lambda_2 \begin{pmatrix} 10 \\ 1 \\ -16 \end{pmatrix}$$

Equating x : $-9 + 3\lambda_1 = 23 + 10\lambda_2$
 $3\lambda_1 - 10\lambda_2 = 32$ (1) ✓

Equating y : $-14 + 2\lambda_1 = -4 + \lambda_2$
 $2\lambda_1 - \lambda_2 = 10$ (2) ✓

Equating z : $-3 + 2\lambda_1 = -27 - 16\lambda_2$
 $2\lambda_1 + 16\lambda_2 = -24$ (3)

(3) - (2) $2\lambda_1 + 16\lambda_2 = -24$ (3)

$2\lambda_1 - \lambda_2 = 10$ (2)

$17\lambda_2 = -34$

$\lambda_2 = -2$

Sub. $\lambda_2 = -2$ in (2)

$2\lambda_1 + 2 = 10$ ✓

$\lambda_1 = 4$

Be careful.
Students must
sub. into an
equation already
used here.

Sub. $\lambda_1 = 4, \lambda_2 = -2$ in (1) to check

sheerness: $3 \times 4 - 10 \times -2 = 32$. ✓ $\rightarrow$ MUST do this!

So point of intersection

$= \begin{pmatrix} -9 \\ -14 \\ -3 \end{pmatrix} + 4 \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$

$= \begin{pmatrix} 3 \\ -6 \\ 5 \end{pmatrix}$

(ii) For 1st line

$x = -9 + 3\lambda_1, y = -14 + 2\lambda_1, z = -3 + 2\lambda_1$

Sub. in to sphere,

$(x-1)^2 + (y+2)^2 + (z-4)^2 = 21$

$(3\lambda_1 - 10)^2 + (2\lambda_1 - 12)^2 + (2\lambda_1 - 7)^2 = 21$

$17\lambda_1^2 - 136\lambda_1 + 293 = 21$

$\lambda^2 - 8\lambda + 16$

$(\lambda_1 - 4)^2 = 0 \Rightarrow \lambda_1 = 4$. Only 1 point of intersection $\rightarrow$ tangent.

Point of intersection $= \begin{pmatrix} -9 \\ -14 \\ -3 \end{pmatrix} + 4 \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ -6 \\ 5 \end{pmatrix}$

Could also show

$\Delta = (-8)^2 - 4 \times 1 \times 16 = 0$

Solutions QMS 2023 Trial: p. 7

Q. (13)(a)(iii) Let direction vector of line $\perp$ to both

$$= \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$$

$$\therefore \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = 0 \quad \& \quad \begin{pmatrix} 10 \\ 1 \\ -16 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = 0$$

$$3x_1 + 2y_1 + 2z_1 = 0 \quad (1)$$

$$10x_1 + y_1 - 16z_1 = 0 \quad (2) \times 2 = (3)$$

$$3x_1 + 2y_1 + 2z_1 = 0 \quad (1) -$$

$$20x_1 + 2y_1 - 32z_1 = 0 \quad (3)$$

$$-17x_1 + 34z_1 = 0 \quad \div -17$$

$$x_1 - 2z_1 = 0$$

$$x_1 = 2z_1$$

Sub. in (1): $6z_1 + 2y_1 + 2z_1 = 0$

$$y_1 = -4z_1$$

Letting $z_1 = 1$, $x_1 = 2$, $y_1 = -4$.

Line $\perp$ to both

$$= \begin{pmatrix} 3 \\ -6 \\ 5 \end{pmatrix} + \lambda_3 \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix}$$

If Line 3 [just found] is radius, then Line 2 is also a tangent as it is $\perp$ to radius.

Finding if $\begin{pmatrix} 3 \\ -6 \\ 5 \end{pmatrix} + \lambda_3 \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix}$ passes through $\begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}$

$$\& x: 3 + 2\lambda_3 = 1, \lambda_3 = -1.$$

Sub. $\lambda_3 = -1$ in Line 3:

$$\begin{pmatrix} 3 \\ -6 \\ 5 \end{pmatrix} - 1 \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} = \underline{\text{sphere centre.}}$$

As Line 2 is $\perp$ to radius at point of contact

on surface of sphere, line 2 = $\begin{pmatrix} 23 \\ -4 \\ -27 \end{pmatrix} + \lambda_2 \begin{pmatrix} 10 \\ 1 \\ -16 \end{pmatrix}$ is ALSO tangent to sphere.

NOTE: Could also use

"or otherwise" & show only 1

point of intersection of Line 2 & sphere.

Solutions QHS 2023 4U Trial p.8

Q.13)(b) Step 1: Show true for $n=2$.

$$\begin{array}{ll} \text{LHS} & \text{RHS} \\ = 7^2 & = 5^2 + 2^2 \\ = 49 & = 29 \end{array}$$

LHS > RHS $\Rightarrow$ True for $n=2$.

Step 2: Assume true for $n=k, k \geq 2$.

i.e. $7^k > 5^k + 2^k$

Step 3: Prove true for $n=k+1$

i.e. RTP: $7^{k+1} > 5^{k+1} + 2^{k+1}, k \geq 2$.

$$\begin{aligned} & \text{LHS} \\ & 7^{k+1} \\ & = 7 \times 7^k \\ & > 7(5^k + 2^k) \text{ [By assumption]} \\ & = (5+2)(5^k + 2^k) \\ & = 5^{k+1} + 5 \times 2^k + 2 \times 5^k + 2^{k+1} \\ & > 5^{k+1} + 2^{k+1} \end{aligned}$$

LHS > RHS.

If it is true for $n=k$, it will be true for $n=k+1$. Hence it will be true $\forall k \in \mathbb{Z}^+ \geq 2$.

[Note: Could also be approached by proving $7^{k+1} - (5^{k+1} + 2^{k+1}) > 0$].

(c.) Assume $\log_7 19$ is rational

i.e. $\log_7 19 = \frac{p}{q}, p, q \in \mathbb{Z}$.

$$\therefore 7^{\frac{p}{q}} = 19$$

$$7^p = 19^q$$

But 7 & 19 are both prime, so a number would have to exist s.t. its prime factors are 7^p & 19^q.

$\therefore$ This is not possible by the fundamental theorem of arithmetic.

$\therefore$ There is a contradiction $\Rightarrow \log_7 19$ is irrational.

Solutions: GHS 2023 Trial p.9

Q. (14) (a) If a, b real

$$(a-b)^2 \geq 0$$

$$a^2 - 2ab + b^2 \geq 0$$

$$a^2 + b^2 \geq 2ab$$

(ii) Using (1) $a^2 + b^2 \geq 2ab$ (1)

$$a^2 + c^2 \geq 2ac$$
 (2)

$$b^2 + c^2 \geq 2bc$$
 (3)

(1) + (2) + (3)

$$2(a^2 + b^2 + c^2) \geq 2(ab + ac + bc)$$

$$a^2 + b^2 + c^2 \geq ab + ac + bc$$

As required.

(iii) If a, b, c are Δ sides:

$$c + b > a \quad \text{or} \quad c + a > b$$

$$c > a - b \quad c > b - a$$

$$\therefore |c| > |a - b|$$

$$c^2 > (a - b)^2$$

Note: IMPORTANT! could also use cosine rule:

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$> a^2 + b^2 - 2ab \text{ as } \cos C \leq 1$$

& if $\cos C = 1$,
no Δ .

(iv) Using (iii) $c^2 > a^2 - 2ab + b^2$ (1)

$$a^2 > b^2 - 2bc + c^2$$
 (2)

$$b^2 > a^2 - 2ac + c^2$$
 (3)

$$(1) + (2) + (3) \quad a^2 + b^2 + c^2 > 2(a^2 + b^2 + c^2) - 2(ab + ac + bc) + 2(ab + ac + bc) - (a^2 + b^2 + c^2)$$

$$2ab + 2ac + 2bc > a^2 + b^2 + c^2$$

as required.

Solutions 2022 GHS 4U Trial p.10

$$Q.(14)(b)(i) I_n = \int_0^{\frac{\pi}{4}} \sin^n x \cdot dx \quad \begin{array}{l} u = \sin^{n-1} x \quad v = -\cos x \\ u' = (n-1)\sin^{n-2} x \cos x \quad v' = \sin x \end{array}$$

$$\text{By } \int u v' \cdot dx = u v - \int v u' \cdot dx$$

$$I_n = \int_0^{\frac{\pi}{4}} \sin^n x \cdot dx = \left[-\cos x \sin^{n-1} x \right]_0^{\frac{\pi}{4}} + \int_0^{\frac{\pi}{4}} \sin^{n-2} x \cos^2 x \cdot dx$$

$$I_n = \left[-\cos \frac{\pi}{4} \left[\sin \frac{\pi}{4} \right]^{n-1} - 0 \right] + (n-1) \int_0^{\frac{\pi}{4}} (1 - \sin^2 x) \sin^{n-2} x \cdot dx$$

$$I_n = \left(-\frac{1}{\sqrt{2}} \right) \times \left(\frac{1}{\sqrt{2}} \right)^{n-1} + (n-1) \int_0^{\frac{\pi}{4}} \sin^{n-2} x \cdot dx - (n-1) \int_0^{\frac{\pi}{4}} \sin^n x \cdot dx$$

$$I_n = -\left(\frac{1}{\sqrt{2}} \right)^n + (n-1) I_{n-2} - (n-1) I_n.$$

$$n I_n = -\left(\frac{1}{\sqrt{2}} \right)^n + (n-1) I_{n-2}$$

$$I_n = -\frac{\left(\frac{1}{\sqrt{2}} \right)^n}{n} + \frac{n-1}{n} I_{n-2}.$$

$$(ii) I_1 = \int_0^{\frac{\pi}{4}} \sin x \cdot dx$$

$$= \left[-\cos x \right]_0^{\frac{\pi}{4}}$$

$$= \left[-\frac{1}{\sqrt{2}} + 1 \right]$$

$$= \left[1 - \frac{1}{\sqrt{2}} \right]$$

$$I_3 = -\frac{\left(\frac{1}{\sqrt{2}} \right)^3}{3} + \frac{2}{3} \left[1 - \frac{1}{\sqrt{2}} \right]$$

$$= \frac{2}{3} - \frac{5}{6\sqrt{2}}$$

$$I_5 = -\frac{\left(\frac{1}{\sqrt{2}} \right)^5}{5} + \frac{4}{5} \left[\frac{2}{3} - \frac{5}{6\sqrt{2}} \right]$$

$$= \frac{8}{15} - \frac{43}{60\sqrt{2}}$$

$$\text{or } I_5 = \frac{64 - 43\sqrt{2}}{120}.$$

Solutions: 2022 GHS 4U Trial p.11

$$Q.(14)(c) \int \frac{x}{\sqrt{2-x}} \cdot dx \times \sqrt{x}$$

$$= \int \frac{x}{\sqrt{2x-x^2}} \cdot dx$$

$$= \int \frac{x-1}{\sqrt{2x-x^2}} \cdot dx + \int \frac{1}{\sqrt{1-(x^2-2x+1)}} \cdot dx$$

$$= -\frac{1}{2} \int \frac{2-2x}{\sqrt{2x-x^2}} \cdot dx + \int \frac{1}{\sqrt{1-(x-1)^2}} \cdot dx$$

$$= -\frac{1}{2} \times 2 \sqrt{2x-x^2} + \sin^{-1}(x-1) + C.$$

$$\left[\text{by } \int f'(x) \cdot [f(x)]^n \cdot dx \right. \\ \left. = \frac{1}{n+1} [f(x)]^{n+1} \right]$$

$$= -\sqrt{2x-x^2} + \sin^{-1}(x-1) + C.$$

$$(d)(i) \text{ By Euler: } \cos n\theta + i \sin n\theta = e^{ni\theta} \quad (1)$$

$$\cos(-n\theta) + i \sin(-n\theta) = e^{-ni\theta}$$

$$\therefore \cos n\theta - i \sin n\theta = e^{-ni\theta} \quad (2)$$

(as cos even, sin odd).

$$(1) - (2): 2i \sin n\theta = e^{ni\theta} - e^{-ni\theta}.$$

$$(ii) \text{ Hence } [2i \sin^5 \theta]^5$$

$$= (e^{i\theta} - e^{-i\theta})^5$$

$$= e^{5i\theta} - 5e^{3i\theta} + 10e^{i\theta} - 10e^{-i\theta} + 5e^{-3i\theta} - e^{-5i\theta}$$

$$= (e^{5i\theta} - e^{-5i\theta}) - 5(e^{3i\theta} - e^{-3i\theta}) + 10(e^{i\theta} - e^{-i\theta})$$

$$\therefore 32i \sin^5 \theta = 2i \sin 5\theta - 10i \sin 3\theta + 20i \sin \theta \quad \div 32i$$

$$\sin^5 \theta = \frac{\sin 5\theta}{16} - \frac{5 \sin 3\theta}{16} + \frac{5 \sin \theta}{8}$$

as required.

Solutions: 2023 GHS Trial p.12

Q. (15)(a) Centre of motion: $0.2\text{m} = C$

(i) Amplitude = 0.05m .

$$\text{Period} = \frac{1}{100} \text{ second.}$$

$$\therefore \frac{2\pi}{n} = \frac{1}{100} \quad \times 100n \quad \times 100n$$

$$\underline{200\pi = n.}$$

$$\therefore \ddot{x} = -(200\pi)^2(x - 0.2).$$

Maximum force will happen at max [or min] accel

i.e. where $x = 0.15$ or $x = 0.25$

[Force will be same, accel = $\pm$].

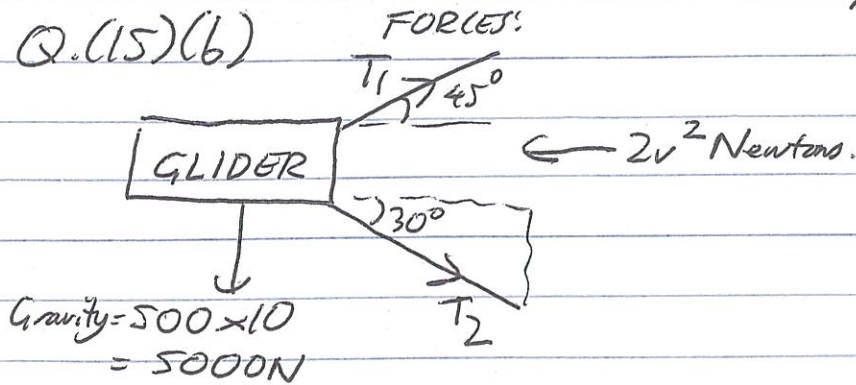
$$\text{At } x = 0.15, a = -(200\pi)^2(0.15 - 0.2).$$

$$\text{By } F = ma,$$

$$F = -(200\pi)^2(-0.05) \times 1.2$$

$$= 23\,687 \text{ Newtons. [nearest Newton].}$$

Q.(15)(b)



(i) Resolving forces vertically

$$T_1 \sin 45^\circ = T_2 \sin 30^\circ + 5000$$

$$\frac{T_1 \sqrt{2}}{2} = \frac{T_2}{2} + 5000$$

$$T_1 \sqrt{2} = T_2 + 10000$$

$$T_1 \sqrt{2} - 10000 = T_2 \text{ as required.}$$

(ii) Resolving horizontally:

$$F = T_1 \cos 45^\circ + T_2 \cos 30^\circ - 2v^2$$

$$500a = \frac{10000 \times \sqrt{2}}{2} + \frac{(10000\sqrt{2} - 10000) \times \sqrt{3}}{2} - 2v^2$$

$$500a = 5000\sqrt{2} + 5000\sqrt{6} - 5000\sqrt{3} - 2v^2$$

$$a = \ddot{x} = 10(\sqrt{2} + \sqrt{6} - \sqrt{3}) - \frac{v^2}{250}$$

Limiting speed is where $\ddot{x} \rightarrow 0$

$$\text{i.e. } \frac{v^2}{250} \rightarrow 10(\sqrt{2} + \sqrt{6} - \sqrt{3})$$

$$\text{Limiting } v = 73 \text{ m/s.}$$

$$(iii) a = v \cdot \frac{dv}{dx} = 10(\sqrt{2} + \sqrt{6} - \sqrt{3}) - \frac{v^2}{250}$$

$$\frac{dv}{dx} = \frac{2500(\sqrt{2} + \sqrt{6} - \sqrt{3}) - v^2}{250v}$$

$$\frac{dx}{dv} = \frac{250v}{2500(\sqrt{2} + \sqrt{6} - \sqrt{3}) - v^2}$$

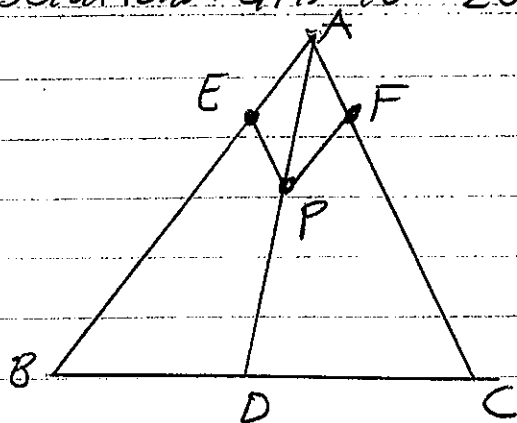
$$x = \int_{50}^{70} \frac{250v}{2500(\sqrt{2} + \sqrt{6} - \sqrt{3}) - v^2} \cdot dv$$

$$= \left[-125 \ln [2500(\sqrt{2} + \sqrt{6} - \sqrt{3}) - v^2] \right]_{50}^{70}$$

$$x = 235.7 \text{ m.}$$

The glide takes
235.7m to get to 70m/s.

Q.15)(c)



$$\begin{aligned} \text{(i)} \quad \vec{BD} &= \lambda \vec{BC} \\ &= \lambda (\vec{BA} + \vec{AC}) \\ &= \lambda (-\vec{p} + \vec{q}) \\ &= \lambda (\vec{q} - \vec{p}) \end{aligned}$$

$$\begin{aligned} \vec{AD} &= \vec{AB} + \vec{BD} \\ &= \vec{p} + \lambda (\vec{q} - \vec{p}) \end{aligned}$$

(ii) Letting $\vec{AP} = k \vec{AD}$

$$\vec{AP} = k [\vec{p} + \lambda \vec{q} - \lambda \vec{p}]$$

$$\vec{AP} = (k - k\lambda) \vec{p} + k\lambda \vec{q} \quad (1)$$

As $\vec{AP} = \vec{AE} + \vec{EP}$

$$\vec{AP} = \frac{1}{3} \vec{p} + \frac{1}{5} \vec{q} \quad (2)$$

$\therefore$ Equating (1) & (2):

$$\frac{1}{3} \vec{p} + \frac{1}{5} \vec{q} = (k - k\lambda) \vec{p} + k\lambda \vec{q}$$

$$\therefore k - k\lambda = \frac{1}{3} \quad \& \quad k\lambda = \frac{1}{5}$$

$$\therefore k - \frac{1}{5} = \frac{1}{3} \Rightarrow k = \frac{8}{15}$$

$$\text{As } k\lambda = \frac{1}{5}, \quad \frac{8}{15} \lambda = \frac{1}{5}$$

$$\begin{aligned} \lambda &= \frac{1}{5} \times \frac{15}{8} \\ &= \frac{3}{8} \end{aligned}$$

$$\therefore \lambda = \frac{3}{8} \quad \& \quad \vec{BD} = \frac{3}{8} \vec{BC}$$

Solutions: GHS 2023 4U Trial p.15

Q. (16)(a) By De Moivre's theorem,
 $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$.

$$\therefore \cos 7\theta + i \sin 7\theta = (\cos \theta + i \sin \theta)^7$$

$$= \cos^7 \theta + 7i \cos^6 \theta \sin \theta - 21 \cos^5 \theta \sin^2 \theta - 35i \cos^4 \theta \sin^3 \theta + 35 \cos^3 \theta \sin^4 \theta + 21i \cos^2 \theta \sin^5 \theta - 7 \cos \theta \sin^6 \theta - i \sin^7 \theta.$$

Equating imaginaries

$$\begin{aligned} \sin 7\theta &= 7 \cos^6 \theta \sin \theta - 35 \cos^4 \theta \sin^3 \theta + 21 \cos^2 \theta \sin^5 \theta - \sin^7 \theta \\ &= 7(1 - \sin^2 \theta)^3 \sin \theta - 35(1 - \sin^2 \theta)^2 \sin^3 \theta + 21(1 - \sin^2 \theta) \sin^5 \theta \\ &= 7 \sin \theta (1 - 3 \sin^2 \theta + 3 \sin^4 \theta - \sin^6 \theta) - 35(1 - 2 \sin^2 \theta + \sin^4 \theta) \sin^3 \theta \\ &\quad + 21 \sin^5 \theta (1 - \sin^2 \theta) - \sin^7 \theta \\ &= 7 \sin \theta - 56 \sin^3 \theta + 112 \sin^5 \theta - 64 \sin^7 \theta \end{aligned}$$

as required.

$$(ii) -\sin 7\theta = 64 \sin^7 \theta - 112 \sin^5 \theta + 56 \sin^3 \theta - 7 \sin \theta$$

$$\text{So if } 64x^7 - 112x^5 + 56x^3 - 7x + 1 = 0$$

$$64x^7 - 112x^5 + 56x^3 - 7x = -1$$

$$-\sin 7\theta = -1$$

$$\& \sin 7\theta = 1$$

$$\text{where } x = \sin \theta.$$

$$\therefore \sin 7\theta = 1$$

$$7\theta = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \frac{13\pi}{2}, \frac{17\pi}{2}, \frac{21\pi}{2}, \frac{25\pi}{2}$$

$$\theta = \frac{\pi}{14}, \frac{5\pi}{14}, \frac{9\pi}{14}, \frac{13\pi}{14}, \frac{17\pi}{14}, \frac{21\pi}{14}, \frac{25\pi}{14}$$

$$\text{Solutions to } 64x^7 - 112x^5 + 56x^3 - 7x + 1 = 0$$

$$\text{are } -1, \sin \frac{\pi}{14}, \sin \frac{5\pi}{14}, \sin \frac{9\pi}{14}, \sin \frac{13\pi}{14}, \sin \frac{17\pi}{14}, \sin \frac{25\pi}{14}.$$

$$(\sin \frac{3\pi}{2})$$

Solutions: GHS 2023 4U Trial p. 16

Q. (16)(a)(iii)

By Product of roots of

$$64x^6 - 112x^5 + 56x^3 - 7x + 1 = 0$$

$$-1 \times \sin \frac{\pi}{14} \times \sin \frac{5\pi}{14} \times \sin \frac{9\pi}{14} \times \sin \frac{13\pi}{14} \times \sin \frac{17\pi}{14} \times \sin \frac{21\pi}{14} = -\frac{1}{64}.$$

$$\div -1 \text{ \& noting } \sin \frac{5\pi}{14} = \sin \frac{9\pi}{14}, \sin \frac{\pi}{14} = \sin \frac{13\pi}{14}$$

$$\text{ \& } \sin \frac{17\pi}{14} = \sin \frac{21\pi}{14}.$$

$$\sin \frac{2\pi}{14} \times \sin \frac{25\pi}{14} \times \sin \frac{21\pi}{14} = \frac{1}{64} \checkmark$$

$$\sin \frac{\pi}{14} \times \sin \frac{5\pi}{14} \times \sin \frac{17\pi}{14} = \pm \frac{1}{8}$$

But as $\sin \frac{\pi}{14}, \sin \frac{5\pi}{14}$ positive \&

$\sin \frac{17\pi}{14}$ negative,

answer is negative.

$$\sin \frac{\pi}{14} \times \sin \frac{5\pi}{14} \times \sin \frac{17\pi}{14} = -\frac{1}{8}.$$

Note: There are a LOT of "or otherwise" methods here!)

Solutions: GMS 4V 2023 Trial p.17

Q. (16) (b) (i)

$$\begin{array}{c} \downarrow \quad \downarrow \\ 0.2 \text{ Mv} \quad 10 \text{ M} \end{array}$$

$$F = Ma = -0.2 \text{ Mv} - 10 \text{ M}$$

$$\therefore a = \frac{dv}{dt} = -0.2v - 10$$

$$\frac{dt}{dv} = -\frac{1}{0.2v+10}$$

$$= -\frac{5}{v+50}$$

$$t = \int_{v_0}^v \frac{-5}{v+50} \cdot dv$$

Note: Could also do $t = \int \frac{-5}{v+50} \cdot dt = -5 \ln(v+50) + C$ & use $v = v_0$ when $t = 0$.

$$= [-5 \ln(v+50)]_{v_0}^v$$

$$= -5 \ln(v+50) + 5 \ln(v_0+50)$$

$$t = -5 \ln \left[\frac{v+50}{v_0+50} \right]$$

Time to max height. $v = 0$

$$t = -5 \ln \left(\frac{50}{v_0+50} \right)$$

Time to max. height. $= 5 \ln \left(\frac{v_0+50}{50} \right)$

(ii) $t = -5 \ln \left[\frac{v+50}{v_0+50} \right]$

$$\therefore e^{-0.2t} = \frac{v+50}{v_0+50}$$

$$(v_0+50)e^{-0.2t} - 50 = v$$

$$x = \int_0^t (v_0+50)e^{-0.2t} - 50 \cdot dt$$

could also do $x = \int (v+50)e^{-0.2t} - 50 \cdot dt$ & use $x = 0$ when $t = 0$.

$$= \left[-5(v_0+50)e^{-0.2t} - 50t \right]_0^t$$

$$= [-5(v_0+50)e^{-0.2t} - 50t] + 5(v_0+50)e^{-0.2 \times 0} + 50 \times 0$$

$$x = -5(v_0+50)e^{-0.2t} - 50t + 5(v_0+50)$$

Solutions: GHS 4U Trial p. 18

Q. (16)(b)(ii) (continued):

Max. height is when $v = 0$ [so $t = 5 \ln \left[\frac{v_0 + 50}{50} \right]$ from (i)].

$$\text{Note: } -0.2t = \ln \left[\frac{50}{v_0 + 50} \right]$$

$$\& e^{-0.2t} = \frac{50}{v_0 + 50}.$$

$$\begin{aligned} \text{So max. height} &= -5[v_0 + 50] \times \frac{50}{v_0 + 50} - 50 \times 5 \ln \left[\frac{v_0 + 50}{50} \right] + 5[v_0 + 50] \\ &= -250 - 250 \ln \left[\frac{v_0 + 50}{50} \right] + 5v_0 + 250 \end{aligned}$$

$$\text{Max. height reached} = \underline{5v_0 - 250 \ln \left(\frac{v_0 + 50}{50} \right)}$$

(16)(b)(iii)

Projectile falling from rest:

↓
10M

0.2MV

↑

$$F = Ma = 10M - 0.2MV$$

$$a = \frac{dv}{dt} = 10 - 0.2v$$

$$\frac{dt}{dv} = \frac{1}{10 - 0.2v}$$

$$= \frac{5}{50 - v}$$

$$t = -5 \int_0^v \frac{-1}{50 - v} dv$$

$$= -5 \left[\ln(50 - v) \right]_0^v$$

$$= -5 \ln \left[\frac{50 - v}{50} \right]$$

$$e^{-0.2t} = \frac{50 - v}{50}$$

$$v = 50 - 50e^{-0.2t}$$

$$x = \int_0^t 50 - 50e^{-0.2t} dt$$

$$= \left[50t + 250e^{-0.2t} \right]_0^t$$

$$= 50t + 250e^{-0.2t} - 250$$

(16)(b)(iv) From (i) & (ii):

As projectile reached a height of $5v_0 - 250 \ln \left[\frac{v_0 + 50}{50} \right]$

after $5 \ln \left[\frac{v_0 + 50}{50} \right]$ seconds,

it will also fall $5v_0 - 250 \ln \left[\frac{v_0 + 50}{50} \right]$ (1)

in $10 - 5 \ln \left[\frac{v_0 + 50}{50} \right]$ seconds.

Using x when projectile is falling

$$x = 50t + 250e^{-0.2t} - 250$$

Note: If $t = 10 - 5 \ln \left[\frac{v_0 + 50}{50} \right]$

$$t = \ln[e^{10}] - 5 \ln \left[\frac{v_0 + 50}{50} \right]$$

$$= -5 \ln[e^{-2}] - 5 \ln \left[\frac{v_0 + 50}{50} \right]$$

$$= -5 \ln \left[\frac{v_0 + 50}{50e^2} \right]$$

$$e^{-0.2t} = \frac{v_0 + 50}{50e^2}$$

$$\& x = 50t + 250e^{-0.2t} - 250$$

$$= 50 \times \left[10 - 5 \ln \left[\frac{v_0 + 50}{50} \right] \right] + 250 \times \frac{(v_0 + 50)}{50e^2} - 250$$

$$= 500 - 250 \ln \left[\frac{v_0 + 50}{50} \right] + \frac{5(v_0 + 50)}{e^2} - 250$$

$$x = 250 + \frac{5(v_0 + 50)}{e^2} - 250 \ln \left[\frac{v_0 + 50}{50} \right] \quad (2)$$

As this is how far projectile has fallen, (1) = (2)

$$5v_0 - 250 \ln \left[\frac{v_0 + 50}{50} \right] = 250 + \frac{5(v_0 + 50)}{e^2} - 250 \ln \left[\frac{v_0 + 50}{50} \right]$$

$$5v_0 = 250 + \frac{5(v_0 + 50)}{e^2}$$

$$5e^2 v_0 = 250e^2 + 5v_0 + 250$$

$$(5e^2 - 5)v_0 = 250e^2 + 250$$

$$v_0 = 65.65 \dots \text{ m/s.}$$